Robust Reinforcement Learning via Leveraging Historically Optimal Policy With Regulation of Performance

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***Abstract*—Most existing adversarial training methods in reinforcement learning (RL) offer limited robustness and remain vulnerable to novel attacks. To address this limitation, an approach that enhances policy robustness by leveraging the historically optimal policy to guide policy optimization and generating diverse adversarial perturbations, termed robust RL via leveraging historically optimal policy with regulation of performance (HORP), is proposed. Unlike other approaches that rely solely on trial-and-error interactions, HORP constructs a guidance value function by simultaneously considering value gaps and policy distribution divergence, thereby focusing on prioritized learning in promising action spaces. It also incorporates an adaptive performance-aware optimization mechanism to trigger timely corrections, preventing the agent from deviating from optimal performance. Furthermore, HORP dynamically modulates perturbation entropy through controlled uncertainty injection, thereby improving the agent's generalized defensive capabilities. Experiments demonstrate that HORP achieves superior performance in most cases regarding both natural performance and robustness against various state attacks.**

***Index Terms*—Adversarial attacks, neural network, policy guidance, policy robustness, reinforcement learning (RL).**

I. Introduction

REINFORCEMENT learning (RL) methods are inherently nonrobust due to their sensitivity to input perturbations and overfitting to specific training environments [1]. However, most current studies assume stationary environments, even though real-world deployments involve stochastic noise and perturbations that can compromise agent control and induce catastrophic performance degradation. To enhance the robustness of RL methods, adversarial training methods have emerged as mainstream solutions [2], most of which focus on developing robustness against worst-case scenarios, typically by training agents against learned optimal attackers within the constrained policy space [3]. Notable works include the alternating training with learned adversaries (ATLA) method [4], which iteratively trains both the agent and its adversary, and its more efficient variant PA-ATLA [5] that introduces a more efficient PA-AD RL attacker. The adversarial training paradigm has been further extended in several directions beyond these foundations. One direction employs game-theoretic methods, such as GRAD, to handle temporally coupled perturbations through strategic interactions with an adversary [6]. Another direction uses bounded rationality curricula, as seen in QARL, to dynamically control the opponent's strength and balance training difficulty [7]. Furthermore, approaches like the ROSE model the adversary's policy distribution to optimize against a diverse set of worst-case scenarios, thereby enhancing the robustness and generalization of the trained policy [8].

Despite these advancements, adversary-based training methods often yield limited improvements in robustness. A primary reason is their sole reliance on trial-and-error interactions with adversaries, which lacks structured guidance. This limitation stems from the fact that agents must simultaneously learn defensive strategies and understand the adversary's behavior patterns, leading to a more complex and potentially inefficient exploration of the policy space. Moreover, these methods face an increased risk of overfitting to specific perturbations as the training progresses, which is detrimental to generalizable robust policy optimization [9]. While recent work such as PROTECTED [10] attempts to address this issue by maintaining a diverse set of defensive policies against various attackers, it significantly increases computational overhead and sample complexity. Consequently, current methods struggle with robustness enhancement inefficiencies and attack-type overfitting, highlighting the need for more adaptive and generalizable solutions.

To address these challenges, we propose a novel method called robust RL via leveraging historically optimal policy with regulation of performance (HORP), which advances iterative adversarial training through an innovative policy guidance mechanism. At its core, HORP implements a policy-guided learning process where each training phase leverages historically optimal policies to guide the development of more robust policies. This guidance mechanism ensures that policies focus on improving adversarial robustness in promising action spaces where improvements over historically optimal policies are feasible and necessary, effectively avoiding redundant learning of well-understood action spaces. HORP incorporates an adaptive optimization mechanism that continuously monitors policy performance during training, dynamically

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adjusting the optimization process based on the performance gap. Furthermore, recognizing that static or diminishing perturbation diversity limits robust generalization, HORP introduces uncertainty into the adversarial perturbations by dynamically regulating the entropy levels of perturbations during training, encouraging continuous exploration of diverse perturbation directions rather than overly targeted defense.

Our key contributions are summarized as follows.

1) We propose a novel policy guidance mechanism that leverages historically optimal policies to improve the adversarial robustness of current policies. By defining the value gap and policy distribution divergence between policies, we design a guidance value function that facilitates targeted policy improvements while ensuring robustness.
2) We develop an adaptive performance-aware optimization method that maintains multiple policies with different replacement strategies. It employs a controller to dynamically adjust policy optimization based on performance gaps, ensuring improvement in policy robustness.
3) We put forward an entropy-regulated training method incorporating controlled uncertainty into adversarial perturbations. By dynamically adjusting the entropy levels of perturbations, we generate diverse adversarial samples, thereby improving the generalization capability of the learned robust policies.

The remainder of this article is organized as follows. Section II reviews robust RL fundamentals and adversarial attacks. Section III introduces HORP in detail, encompassing policy-guided learning, adaptive performance-aware optimization, and entropy-regulated adversarial training. Section IV evaluates HORP's empirical performance, and Section V presents the conclusions.

II. Related Work

A. Robust RL

Robustness in RL has multiple interpretations and covers various aspects, such as action perturbations, reward corruptions, and dynamics uncertainty [11]. Researchers have developed defense strategies against action perturbations, ranging from noise injection [12] to robust optimization techniques [13]. Recent advances include game-theoretic methods like GRAD [6] that handles temporally coupled perturbations, and QARL [7] that employs bounded rationality curricula to balance training difficulty.

A fundamental challenge in robust RL is accurately estimating Markov decision processes (MDPs) model parameters. To address this challenge, robust MDPs (RMDPs) [14] have emerged as a promising method for managing model ambiguity. The RMDP method, which considers worst-case perturbations in transition probabilities, has been successfully extended to distributional settings [15] and partially observed MDPs [16]. Constrained RMDPs (CRMDPs) formulate a max–min optimization problem to find the best policy under worst-case transition dynamics within a predefined uncertainty set [17]. Subsequent works extended CRMDPs to offline learning [18] with coverage assumptions such as robust single-policy clipped concentrability [14] and uniformly well coverage [19]. In parallel, the regularized RMDP (RRMDP) method replaces hard constraints with a divergence-based penalty, as studied in the offline setting by Zhang et al. [20]. Both lines have been extended to function approximation by Panaganti et al. [11] and Tang et al. [21] in hybrid and purely offline scenarios. The pursuit of sample efficiency has seen progress with near-optimal complexity for general L_p norms [22], which provides a theoretical characterization of uncertainty sets, in contrast to HORP's practical method of managing uncertainty through dynamically regulated perturbation entropy. Further extensions include promoting robustness through flat reward maxima in policy space [23] via implicit guidance toward flat parameter regions, while HORP offers explicit optimization direction through policy value gap and policy distribution divergence, and addressing dynamics mismatch in Sim2Real transfer [24] through precise physical dynamics modeling, whereas HORP establishes direct policy-level guidance via historically optimal policies. Complementing these are practical methods for online settings [25], [26] and offline settings [27].

Our work primarily focuses on the observational robustness of RL agents under state adversarial attacks [28]. This area has witnessed significant developments through various methods, including regularization techniques [29], gradient-based attack-driven methods [30], and alternating training strategies [4], [5]. Recent research has further diversified into several promising directions. Provable robustness methods, such as certified radius maximization [31] and randomized smoothing [32], aim to guarantee performance under bounded perturbations. Representation learning methods, including RobustZero [33] for consistent hidden states and CAR-DQN [34] for Bellman infinity-error optimization, seek to learn more stable state encodings. Other strategies address specific challenges: BAT handles environmental perturbations [35], ARA-RL targets temporally coupled disturbances [36], and ROSE optimizes against worst-case adversary distributions [8]. Additionally, PROTECTED [10] enhances performance against nonworst-case attacks through iterative discovery of nondominated policies. While these methods have made considerable progress, they often rely on trial-and-error without structured guidance, resulting in suboptimal policy exploration. In contrast, HORP introduces systematic policy guidance that leverages historically optimal policies and maintains perturbation diversity, addressing these limitations directly. A detailed comparative analysis of Robust RL methods is provided in Section S1 of the supplementary material.

B. Adversarial Attacks on RL

State-space adversarial attacks on RL policies have evolved to include a range of sophisticated methods. Notable among these are advanced pixel-based attacks [29] and critic-independent strategies such as RS and MAD [37]. Sun et al. [5] further advance this field by developing the PA-AD RL attacker within their PAMDP method, achieving improved efficiency. Recent attack methods have emphasized stealth and efficiency, such as the illusory attacks proposed by

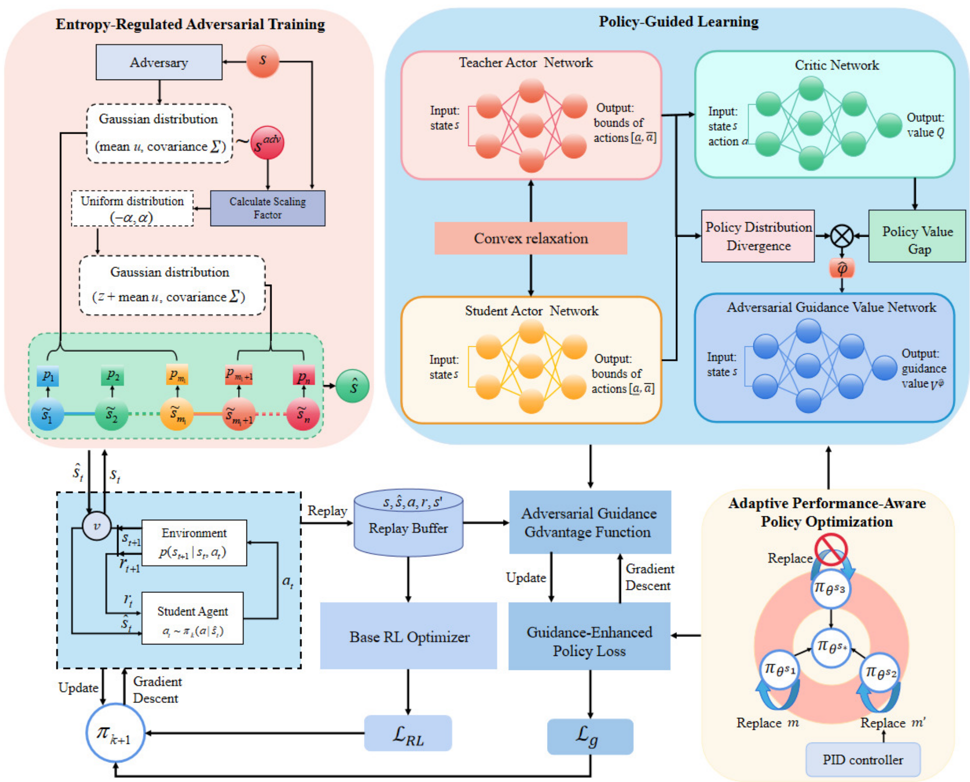

Fig. 1. Schematic of the HORP method illustrating the integration of policy-guided learning, adaptive performance-aware policy optimization, and entropy-regulated adversarial training into the iterative robustness enhancement method.

Franzmeyer et al. [38] that are constrained by information-theoretic detectability.

Beyond state-based attacks, RL agents have shown vulnerability to action space perturbations [39]. Notable contributions include the MAS and LAS attacks [40], which target cumulative reward minimization under different constraint structures, and action poisoning attacks [41] that manipulate the agent's action selection process. In multiagent settings, adversarial policies have been shown to induce malicious state sequences through strategic behavior manipulation rather than direct state modification [42].

Reward poisoning attacks present another critical threat vector, where adversaries subtly corrupt reward signals to steer agents toward undesirable policies [43]. Banihashem et al. [44] analyze cost-effective targeted reward manipulation, while Xu et al. [45] demonstrate fundamental vulnerabilities through their adversarial MDP method, enabling closed box reward poisoning. Concurrently, model extraction attacks compound these risks, as identified by Ilahi et al. [46], who highlight dual threats of adversarial behavior induction and policy theft. Comprehensive surveys like Standen et al. [2] have systematically categorized such adversarial attacks, providing a unified framework for understanding the threat landscape.

Despite these advances, current attack methods suffer from diminishing stochasticity as training progresses, which reduces exploratory effectiveness and ultimately limits their capacity to enhance policy robustness through adversarial training. This inherent limitation underscores the need for more sophisticated perturbation mechanisms that maintain exploratory diversity throughout the learning process, a gap that our entropy-regulated adversarial training specifically addresses.

III. Methodology

This section outlines the HORP method for improving policy robustness via iterative adversarial training. HORP departs from methods reliant solely on trial-and-error by advancing three novel components: 1) a policy guidance mechanism that leverages historical optimal policies to focus learning; 2) an adaptive performance-aware component to prevent optimization deviation; and 3) an entropy-regulated adversarial trainer that promotes generalization through diverse perturbations. These integrated components collectively enhance robustness, as illustrated in Fig. 1.

A. State Space Threat Model

1) Notations: We consider a finite-horizon Markov decision process defined by the tuple $(\mathcal{S}, \mathcal{A}, \mathcal{P}, r, \gamma, T, \mu_0)$, where $\mathcal{S}$ and $\mathcal{A}$ are compact state and action spaces; $\mathcal{P} : \mathcal{S} \times \mathcal{A} \times \mathcal{S} \to [0, 1]$ is the transition kernel, with $p(s'|s, a)$ denoting the probability of transitioning from state s to s' under action a; the reward function is given by $r : \mathcal{S} \times \mathcal{A} \to \mathbb{R}$; $\gamma \in [0, 1)$ is the discount factor; T is the time horizon; and $\mu_0 \in \Delta(\mathcal{S})$ represents the initial state distribution. A stationary policy $\pi : \mathcal{S} \to \mathcal{P}(\mathcal{A})$ maps states to distributions over actions. Let Π denote the set of deterministic stationary policies, and $\mathcal{P}(\Pi)$ the set of stochastic policies. The goal of RL is to maximize the expected discounted return. We define the value function and Q-function as follows:

$$V^{\pi}(s) = \mathbb{E}_{\tau\sim\pi}\left[\sum_{t=0}^{T}\gamma^t r(s_t, a_t)\,|s_0 = s\right]$$
$$Q^{\pi}(s,a) = \mathbb{E}_{\tau\sim\pi}\left[\sum_{t=0}^{T}\gamma^t r(s_t, a_t)\,|s_0 = s, a_0 = a\right] \quad (1)$$

where the expectation is taken over trajectories $\tau = (s_0, a_0, \ldots, s_T, a_T)$.

2) Threat Model: Following common settings in adversarial RL [4], [10], we consider an adversary that perturbs the observed state. The adversary is modeled as a function $\nu : \mathcal{S} \to \mathcal{S}$, with $\nu(s) \in \mathcal{B}(s) \subset \mathcal{S}$ constrained to a bounded set of admissible perturbations. Randomized adversaries are also considered, with strategy space $\Delta(\mathcal{V}^{\text{det}})$. Under this threat model, the perturbed value and Q-function are defined as follows:

$$\hat{V}^{\pi\circ\nu}(s) = \mathbb{E}_{\substack{\hat{s}\sim\nu(s)\\ a\sim\pi(\hat{s})\\ s'\sim p(\cdot|s,a)}}\left[\sum_t \gamma^t r(s_t, a_t)\,|s_0 = s\right]$$
$$\hat{Q}^{\pi\circ\nu}(s,a) = \mathbb{E}_{\substack{\hat{s}\sim\nu(s)\\ a\sim\pi(\hat{s})\\ s'\sim p(\cdot|s,a)}}\left[\sum_t \gamma^t r(s_t, a_t)\,|s_0 = s, a_0 = a\right]. \quad (2)$$

B. Policy-Guided Learning

The core innovation of HORP is a policy guidance mechanism that establishes dynamic knowledge transfer between policy iterations. We formalize this as a teacher–student method: the historically optimal policy accumulated from previous iterations serves as the teacher policy π_{θ^T}, while the current policy under training is the student π_{θ^S}, with θ^T and θ^S denoting their respective parameters. The method is initialized with a teacher policy π_{θ^T} possessing limited robustness, typically obtained from prior robust RL methods such as ATLA [4] or WocaR [3].

The training proceeds through sequential stages. In each stage, three student policies $\{\pi_{\theta^{S_i}}\}_{i=1}^{3}$, whose distinct roles are detailed in Section III-C, are trained under adversarial conditions, guided by the teacher π_{θ^T}. Each stage concludes with an independent evaluation of all policies. If the optimal student policy surpasses the teacher's performance, it becomes the new teacher for the subsequent stage. The training process terminates when the teacher policy remains unchanged for $K_{\min}$ consecutive iterations, indicating convergence. To formally characterize the guidance from π_{θ^T} to π_{θ^S}, we propose two fundamental metrics that underpin our guidance mechanism: the Policy Value Gap and the Policy Distribution Divergence.

Definition 1 (Policy Value Gap): For any state $s \in \mathcal{S}$ and action $a \in \mathcal{A}$, the Policy Value Gap $Q_{\theta^S|\theta^T}(s,a)$ quantifies the advantage of the student's action over the teacher's achievable value

$$Q_{\theta^S|\theta^T}(s,a) = Q^{\pi_{\theta^S}}(s,a) - \max_{\hat{a}\in[\underline{a},\overline{a}]} Q^{\pi_{\theta^S}}(s,\hat{a}) \quad (3)$$

where the action bounds $[\underline{a},\overline{a}]$ are derived from the teacher's output range under state perturbations within $\mathcal{B}(s)$

$$\underline{a} = \min_{\hat{s}\in\mathcal{B}(s)} \pi_{\theta^T}(\hat{s}), \quad \overline{a} = \max_{\hat{s}\in\mathcal{B}(s)} \pi_{\theta^T}(\hat{s}). \quad (4)$$

The bounds are computed using existing neural network convex relaxation techniques [3], and the maximizing $\hat{a}$ is obtained via gradient ascent, following established practices [45].

Definition 2 (Adversarially Perturbed Policy Value Gap): Under the adversary ν, the gap becomes

$$\hat{Q}_{(\theta^S|\theta^T)\circ\nu}(s,a) = \hat{Q}^{\pi_{\theta^S}\circ\nu}(s,a) - \max_{\hat{a}\in[\underline{a},\overline{a}]} \hat{Q}^{\pi_{\theta^S}\circ\nu}(s,\hat{a}). \quad (5)$$

Lemma 1 (Perturbation Bound for Policy Value Gap): The difference between the standard and adversarially perturbed Policy Value Gap is bounded

$$\max_{\nu}\max_{s,a}\left|Q_{\theta^S|\theta^T}(s,a) - \hat{Q}_{(\theta^S|\theta^T)\circ\nu}(s,a)\right|$$
$$\leq \alpha \max_{s}\max_{\hat{s}\in\mathcal{B}(s)} \mathrm{D_{TV}}\left(\pi_{\theta^S}(\cdot|s), \pi_{\theta^S}(\cdot|\hat{s})\right) \quad (6)$$

where $\alpha = 4\gamma[1 + \gamma/((1-\gamma)^2)]\max_{s,a}|r(s,a)|$. The proof is given in Section S2.1 of the supplementary material.

Definition 3 (Policy Distribution Divergence): For any state $s \in \mathcal{S}$ and action $a \in \mathcal{A}$, the Policy Distribution Divergence $\mathcal{D}_{\theta^S|\theta^T}(s,a)$ quantifies the difference in action selection probability between the student and teacher policies under state perturbations

$$\mathcal{D}_{\theta^S|\theta^T}(s,a) = \underline{\pi_{\theta^S}}(a \mid s) - \overline{\pi_{\theta^T}}(a \mid s). \quad (7)$$

For continuous action spaces with diagonal Gaussian policies and state-independent covariance Σ, we derive the probability bounds as follows:

$$\underline{\pi_{\theta^S}}(a \mid s) = \frac{e^{-\overline{d}(a,\underline{a_S},\overline{a_S})/2}}{\left((2\pi)^i \det\Sigma\right)^{0.5}}$$
$$\overline{\pi_{\theta^T}}(a \mid s) = \frac{e^{-\underline{d}(a,\underline{a_T},\overline{a_T})/2}}{\left((2\pi)^i \det\Sigma\right)^{0.5}} \quad (8)$$

where the distance functions are defined as follows;

$$\overline{d}\left(a,\underline{a},\overline{a}\right) = \max_{\hat{a}\in[\underline{a},\overline{a}]} (a-\hat{a})^T\Sigma^{-1}(a-\hat{a})$$
$$\underline{d}\left(a,\underline{a},\overline{a}\right) = \min_{\hat{a}\in[\underline{a},\overline{a}]} (a-\hat{a})^T\Sigma^{-1}(a-\hat{a}). \quad (9)$$

Here, $\underline{a_S}, \overline{a_S}$ and $\underline{a_T}, \overline{a_T}$ denote the bounds of mean action outputs for student and teacher policies, respectively, obtained via convex relaxation techniques [3]. The constant i represents the action space dimension.

Property 1 (Boundedness of Policy Distribution Divergence): For any state $s \in \mathcal{S}$ and action $a \in \mathcal{A}$, the Policy Distribution Divergence is bounded. Formally, there exists a constant $C > 0$ such that

$$\left|\mathcal{D}_{\theta^S|\theta^T}(s,a)\right| \leq C \quad (10)$$

where $C = ((2\pi)^i \det\Sigma)^{-0.5}$ and i is the action space dimension. The proof is given in Section S2.2 of the supplementary material.

To integrate the Policy Value Gap (Definition 1) and the Policy Distribution Divergence (Definition 3) into a unified learning signal, we define the guidance reward function as follows:

Definition 4 (Guidance Reward): The guidance reward function $\varphi_{\theta^S|\theta^T} : \mathcal{S}\times\mathcal{A}\to\mathbb{R}$ is defined as follows:

$$\varphi_{\theta^S|\theta^T}(s,a) = \operatorname{sgn}\left(Q_{\theta^S|\theta^T}(s,a)\right)$$

$$\cdot \max\left(0, Q_{\theta^S|\theta^T}(s,a)\,\mathcal{D}_{\theta^S|\theta^T}(s,a)\right) \tag{11}$$

where $\mathrm{sgn}(\cdot)$ is the sign function.

As a key innovation of our method, this reward function strategically combines value and distribution information to prioritize learning. It actively encourages the student to explore promising actions ($Q_{\theta^S|\theta^T}(s,a) > 0$) in underexplored regions ($\mathcal{D}_{\theta^S|\theta^T}(s,a) > 0$) by providing positive rewards, while discouraging suboptimal behaviors ($Q_{\theta^S|\theta^T}(s,a) < 0$) in overexplored areas ($\mathcal{D}_{\theta^S|\theta^T}(s,a) < 0$) through negative rewards. When actions offer no meaningful improvement potential ($\varphi_{\theta^S|\theta^T}(s,a) = 0$), the reward neutralizes to zero, ensuring focused exploration where enhancement over the teacher is both feasible and impactful.

Definition 5 (Adversarial Guidance Reward): Under the adversary ν, the guidance reward becomes

$$\hat{\varphi}_{(\theta^S|\theta^T)\circ\nu}(s,a) = \mathrm{sgn}\left(\hat{Q}_{(\theta^S|\theta^T)\circ\nu}(s,a)\right) \cdot \max\left(0, \hat{Q}_{(\theta^S|\theta^T)\circ\nu}(s,a)\,\mathcal{D}_{\theta^S|\theta^T}(s,a)\right). \tag{12}$$

Property 2 (Magnitude Bound of Guidance Reward): Both standard and adversarial guidance rewards are uniformly bounded

$$\max\left\{|\varphi_{\theta^S|\theta^T}(s,a)|, |\hat{\varphi}_{(\theta^S|\theta^T)\circ\nu}(s,a)|\right\} \leq \frac{2C\max_{s,a}|r(s,a)|}{1-\gamma} \tag{13}$$

where $C = ((2\pi)^i \det\Sigma)^{-0.5}$ from Property 1. The proof is given in Section S2.3 of the supplementary material.

Theorem 1 (Guidance Reward Stability under Adversarial Perturbations): The guidance reward remains stable against adversarial attacks, with bounded deviation

$$\max_{s,a}\left|\varphi_{\theta^S|\theta^T}(s,a) - \hat{\varphi}_{(\theta^S|\theta^T)\circ\nu}(s,a)\right| \leq C\alpha \max_{s}\max_{\hat{s}\in\mathcal{B}(s)} \mathrm{D_{TV}}\left(\pi_{\theta^S}(\cdot|s), \pi_{\theta^S}(\cdot|\hat{s})\right) \tag{14}$$

where $\alpha = 4\gamma[1 + \gamma/((1-\gamma)^2)]\max_{s,a}|r(s,a)|$ from Lemma 1. The proof is given in Section S2.4 of the supplementary material.

To evaluate the long-term effectiveness of the guidance mechanism, we define the guidance value function $V^{\varphi}_{\theta^S|\theta^T}$ as the expected cumulative guidance rewards

$$V^{\varphi}_{\theta^S|\theta^T}(s) = \mathbb{E}_{\tau\sim\pi_{\theta^S}, s_0=s}\left[\sum_{t=0}^{T}\gamma^t\varphi_{\theta^S|\theta^T}(s_t,a_t)\right] \tag{15}$$

where $\tau = (s_0, a_0, s_1, a_1, \ldots,)$ denotes trajectories under π_{θ^S}.

Building upon this definition, we establish the theoretical foundation of our guidance method through the following results:

Theorem 2 (Bellman Contraction for Guidance Value): The guidance value function satisfies the Bellman equation $V^{\varphi}_{\theta^S|\theta^T} = \mathcal{T}V^{\varphi}_{\theta^S|\theta^T}$, where the Bellman operator $\mathcal{T}$ is defined as follows:

$$(\mathcal{T}V)(s) = \sum_{a\in\mathcal{A}}\pi_{\theta^S}(a|s)\sum_{s'\in\mathcal{S}}p\left(s'|s,a\right)\left[\varphi_{\theta^S|\theta^T}(s,a) + \gamma V\left(s'\right)\right]. \tag{16}$$

The operator $\mathcal{T}$ is a contraction mapping under the supremum norm and admits a unique fixed point. The proof is given in Section S2.5 of the supplementary material.

To address the adversarial setting, we extend our analysis to the perturbed guidance value:

Theorem 3 (Bellman Equation for Adversarial Guidance Value): For fixed student policy π_{θ^S} and adversary ν, the adversarial guidance value function satisfies

$$V^{\hat{\varphi}}_{(\theta^S|\theta^T)\circ\nu}(s) = \sum_{a\in\mathcal{A}}\pi_{\theta^S}(a|\nu(s))\sum_{s'\in\mathcal{S}}p\left(s'|s,a\right)\left[\hat{\varphi}_{(\theta^S|\theta^T)\circ\nu}(s,a) + \gamma V^{\hat{\varphi}}_{(\theta^S|\theta^T)\circ\nu}\left(s'\right)\right]. \tag{17}$$

The proof is given in Section S2.6 of the supplementary material.

Building upon the adversarial guidance value, we further consider the worst-case scenario to establish the robustness guarantees of our method.

Definition 6 (Optimal Adversarial Guidance Value): For a fixed student policy π_{θ^S}, the optimal adversarial guidance value under the worst-case adversary is defined as follows:

$$V^{\hat{\varphi}}_{(\theta^S|\theta^T)\circ\nu^*}(s) = \min_{\nu\in\mathcal{V}} V^{\hat{\varphi}}_{(\theta^S|\theta^T)\circ\nu}(s). \tag{18}$$

Theorem 4 (Robustness Bound for Guidance Value under Optimal Adversary): The discrepancy between the standard guidance value and its optimal adversarial counterpart is bounded by

$$\max_{s\in\mathcal{S}}\left|V^{\varphi}_{\theta^S|\theta^T}(s) - V^{\hat{\varphi}}_{(\theta^S|\theta^T)\circ\nu^*}(s)\right| \leq \frac{L}{1-\gamma}\cdot\max_{s\in\mathcal{S},\hat{s}\in\mathcal{B}(s)}\mathrm{D_{TV}}\left(\pi_{\theta^S}(\cdot|s), \pi_{\theta^S}(\cdot|\hat{s})\right) \tag{19}$$

where $L = (4C\max_{s,a}|r(s,a)|)/((1-\gamma)^2) + C\alpha$, with constants C and α from Property 1 and Lemma 1, respectively. The proof is given in Section S2.7 of the supplementary material.

To implement the guidance mechanism, we integrate it with conventional actor–critic methods within a practical learning method. Our core contribution is the use of historically optimal policies to construct specialized guidance signals for policy optimization. During training, policy updates utilize experience tuples $\{s_t, \hat{s}_t, a_t, r_t, s_{t+1}\}$ collected under adversarial conditions, where $\hat{s}_t \sim \nu(s_t)$ and $a_t \sim \pi_{\theta^S_k}(\cdot|\hat{s}_t)$. Here, $\pi_{\theta^S_k}$ denotes the behavior policy (the policy parameters from the previous iteration) that generated the data. The agent minimizes a guidance-enhanced policy loss to favor actions with higher adversarial guidance value

$$\mathcal{L}_{\mathrm{g}}\left(\pi_{\theta^S}\circ\nu\right) = -\frac{1}{N}\sum_{t=0}^{N}\sum_{a\in\mathcal{A}}\left[\frac{\pi_{\theta^S}(a|\hat{s}_t)}{\pi_{\theta^S_k}(a|\hat{s}_t)}A^{\hat{\varphi}}_{(\theta^S_k|\theta^T)\circ\nu}(s_t,a)\right] \tag{20}$$

where the adversarial guidance advantage function is

$$A^{\hat{\varphi}}_{(\theta^S_k|\theta^T)\circ\nu}(s_t,a) = \hat{\varphi}_{(\theta^S_k|\theta^T)\circ\nu}(s_t,a) + \gamma V^{\hat{\varphi}}_{(\theta^S_k|\theta^T)\circ\nu}\left(s_{t+1}\right) - V^{\hat{\varphi}}_{(\theta^S_k|\theta^T)\circ\nu}(s_t). \tag{21}$$

The HORP combines this guidance mechanism with standard RL methods through a composite loss

$$\mathcal{L}_{\mathrm{HORP}}\left(\pi_{\theta^S}\circ\nu\right) = \mathcal{L}_{\mathrm{RL}}\left(\pi_{\theta^S}\circ\nu\right) + \lambda\mathcal{L}_{\mathrm{g}}\left(\pi_{\theta^S}\circ\nu\right) \tag{22}$$

where $\mathcal{L}_{\mathrm{RL}}(\pi_{\theta^S} \circ \nu)$ is the loss of the base RL method. For theoretical analysis, we employ the PPO-penalty formulation [47] as the base method

$$\mathcal{L}_{\mathrm{RL}}(\pi_{\theta^S} \circ \nu) = -\mathbb{E}_{\tau \sim \pi_{\theta_k^S} \circ \nu}\left[\frac{\pi_{\theta^S}(a_t|\hat{s}_t)}{\pi_{\theta_k^S}(a_t|\hat{s}_t)} A_{\pi_{\theta_k^S} \circ \nu}(s_t, a_t)\right] + \beta \mathbb{E}_{s \sim \rho^{\pi_{\theta_k^S} \circ \nu}}\left[D_{\mathrm{KL}}\left(\pi_{\theta^S}(\cdot|\nu(s)) \| \pi_{\theta_k^S}(\cdot|\nu(s))\right)\right] \tag{23}$$

with $A_{\pi_{\theta_k^S} \circ \nu}(s_t, a_t) = \hat{Q}^{\pi_{\theta_k^S} \circ \nu}(s_t, a_t) - \hat{V}^{\pi_{\theta_k^S} \circ \nu}(s_t)$ denoting the environmental advantage function.

We now analyze the policy improvement guarantee of HORP. Let π_{θ^S} be the current student policy and $\pi_{\theta'^S}$ be the updated policy after minimizing $\mathcal{L}_{\mathrm{HORP}}$. Define the expected environment return as $J(\pi \circ \nu) = \mathbb{E}_{s \sim \mu_0}[\hat{V}^{\pi \circ \nu}(s)]$, where $\hat{V}^{\pi \circ \nu}(s)$ is the perturbed value function for environmental rewards as defined in Equation 2.

Theorem 5 (Policy Improvement Guarantee under Adversarial Perturbations): A sufficient condition for policy improvement $J(\pi_{\theta'^S} \circ \nu) \geq J(\pi_{\theta^S} \circ \nu)$ is given by

$$\beta \mathbb{E}_{s \sim \rho^{\pi_{\theta^S} \circ \nu}}[D_{\mathrm{KL}}] - C_{\mathrm{env}} \max_s D_{\mathrm{KL}} \geq \lambda K_1 \tag{24}$$

where $D_{\mathrm{KL}} = D_{\mathrm{KL}}(\pi_{\theta^S}(\cdot|\nu(s)) \| \pi_{\theta'^S}(\cdot|\nu(s)))$ denotes the KL divergence, $C_{\mathrm{env}} = (4\gamma \max_{s,a} |A_{\pi_{\theta^S} \circ \nu}(s,a)|)/((1-\gamma)^2)$ is a constant derived from the performance difference lemma [48], and $K_1 = (4C \max_{s,a} |r(s,a)|)/((1-\gamma)^2)$ is a constant parameter. The proof is given in Section S2.8 of the supplementary material.

This theoretical bound offers practical guidance for hyperparameter selection: increasing the KL penalty β strengthens the improvement guarantee, while the guidance weight λ should be carefully calibrated against the magnitude of policy updates. The condition is more readily satisfied when the expected KL divergence significantly exceeds its maximum value across states.

HORP alternates between updating the student policy π_{θ^S} to minimize $\mathcal{L}_{\mathrm{HORP}}$ and updating the adversary ν_ϕ to maximize $\mathcal{L}_{\mathrm{RL}}$ (i.e., minimize the environmental return). Under standard assumptions (see Section S2.9 of the supplementary material for details), this leads to the following convergence result.

Theorem 6 (Convergence of HORP): Let ϕ denote the parameters of the adversary ν_ϕ. The HORP method converges to a stationary point where

$$\lim_{k \to \infty} \mathbb{E}\left[\left\|\nabla_{\theta^S} \mathcal{L}_{\mathrm{HORP}}(\theta_k^S, \phi_k)\right\|^2 + \left\|\nabla_\phi \mathcal{L}_{\mathrm{RL}}(\theta_k^S, \phi_k)\right\|^2\right] = 0. \tag{25}$$

Moreover, after K_2 iterations

$$\min_{0 \leq k \leq K_2} \mathbb{E}\left[\left\|\nabla_{\theta^S} \mathcal{L}_{\mathrm{HORP}}(\theta_k^S, \phi_k)\right\|^2 + \left\|\nabla_\phi \mathcal{L}_{\mathrm{RL}}(\theta_k^S, \phi_k)\right\|^2\right] \leq \frac{M}{\sqrt{K_2}} \tag{26}$$

where M is a constant. The detailed assumptions and proof are provided in Section S2.9 of the supplementary material.

C. Adaptive Performance-Aware Policy Optimization

To prevent deviation from optimal learning trajectories, we propose a novel adaptive optimization mechanism that dynamically regulates policy updates based on performance feedback. Our key innovation lies in the coordinated use of three student policies with distinct reset strategies, effectively balancing policy refinement with exploratory learning.

During each iteration of M environment steps, we maintain three student policies $\{\pi_{\theta^{S_i}}\}_{i=1}^3$ with specialized roles.

1) π_{θ^1} resets to the best student policy π_{θ^*} every fixed m steps, providing a stable performance baseline.
2) π_{θ^2} employs a dynamically adjusted reset interval m' computed using an existing proportional–integral–derivative (PID) controller [49] based on real-time performance gaps.
3) π_{θ^3} pursues continuous learning without resets, enabling sustained exploration.

At each policy replacement step, we update π_{θ^*} by comparing the perturbed value functions $\hat{V}^\pi(s)$ of all three student policies, selecting the highest-performing one as the new π_{θ^*}.

Definition 7 (Student Performance Gap): For the student policy π_{θ^2}, the Student Performance Gap $\mathcal{G}_j$ at the jth control step quantifies the difference in the perturbed value function between the best student policies π_{θ^*} and π_{θ^2}

$$\mathcal{G}_j = \hat{V}^{\pi_{\theta^*} \circ \nu}(s) - \hat{V}^{\pi_{\theta^2} \circ \nu}(s) \tag{27}$$

where $\hat{V}^\pi(s)$ is the perturbed value function from 2.

The PID controller computes output $o(j)$ as follows:

$$o(j) = K_p \cdot \mathcal{G}_j + K_i \cdot \sum_{n=0}^{j} \mathcal{G}_n + K_d \cdot (\mathcal{G}_j - \mathcal{G}_{j-1}) \tag{28}$$

with K_p, K_i, and K_d representing proportional, integral, and derivative gains, respectively.

The parameter selection is principled by their distinct regulatory roles: K_p provides immediate correction with the dominant value to ensure responsive performance; K_i addresses persistent errors while avoiding wind-up through a relatively small setting; and K_d mitigates overshoot without amplifying noise via a minor gain configuration.

This parameter combination enables adaptive response characteristics: during minor performance declines, K_p and K_i facilitate smooth and gradual adjustments, while under severe performance degradation, the combined action of K_p and K_d ensures rapid corrective recovery. The specific coefficients were determined based on their empirical relative magnitudes ($K_p > K_i > K_d$) and then finely tuned via grid search. A comprehensive sensitivity analysis in Section S3.1 of the supplementary material validates this configuration and demonstrates its influence on overall performance and robustness.

The replacement interval for π_{θ^2} is then determined by

$$m' = \frac{m}{o(j)}. \tag{29}$$

This adaptive mechanism ensures more frequent resets when performance degrades and longer intervals during successful exploration.

At iteration end, π_{θ^*} replaces the teacher policy π_{θ^T} if it demonstrates superior performance under adversarial perturbations, completing the knowledge transfer cycle as detailed in Algorithm 1.

Algorithm 1 Robust RL via Leveraging Historically Optimal Policy With Regulation of Performance

Input: $K_{\min}$, N, T, m, $V^{\varphi}_{\theta^{S_1}|\theta^T}$, $V^{\varphi}_{\theta^{S_2}|\theta^T}$, $V^{\varphi}_{\theta^{S_3}|\theta^T}$, $\hat{V}$
Initialize π_{θ^T}
while π_{θ^T} has not improved for $K_{\min}$ iterations **do**
  Initialize $\{\pi_{\theta^{S_i}}\}_{i=1}^{3}$, $m' = m$, $\pi_{\theta^*} = \pi_{\theta^{S_3}}$
  for $n = 1$ to N **do**
    for each student policy $\pi_{\theta^{S_i}}$ **do**
      Collect a set of trajectories $\mathcal{D} = \{\tau_k\}$ using $\pi_{\theta^{S_i}}$ under adversarial perturbations from Algorithm 2, where each $\tau_k = \{(s_{k,t}, \hat{s}_{k,t}, a_{k,t}, r_{k,t}, s_{k,t+1})\}_{t=0}^{|\tau_k|}$
      Compute action bounds $[\underline{a_S}_{k,t}, \overline{a_S}_{k,t}]$ and $[\underline{a_T}_{k,t}, \overline{a_T}_{k,t}]$ via convex relaxation techniques for each state $s_{k,t}$
      Calculate Policy Value Gap $\hat{Q}_{(\theta^{S_i}|\theta^T)\circ\nu}(s_{k,t}, a_{k,t})$ and Policy Distribution Divergence $\mathcal{D}_{\theta^{S_i}|\theta^T}(s_{k,t}, a_{k,t})$
      Calculate adversarial guidance reward $\hat{\varphi}_{(\theta^{S_i}|\theta^T)\circ\nu}$ for each step t in every episode k
      Update policy parameters after minimizing $\mathcal{L}_{\text{HORP}}$
    end for
    if $n \bmod m = 0$ **then**
      $\pi_{\theta^*} = \arg\max_{\pi\in\{\pi_{\theta^*}, \pi_{\theta^{S_1}}, \pi_{\theta^{S_2}}, \pi_{\theta^{S_3}}\}} \hat{V}^{\pi}$
      Replace $\pi_{\theta^{S_1}} = \pi_{\theta^*}$
    end if
    if $n \bmod m' = 0$ **then**
      $\pi_{\theta^*} = \arg\max_{\pi\in\{\pi_{\theta^*}, \pi_{\theta^{S_1}}, \pi_{\theta^{S_2}}, \pi_{\theta^{S_3}}\}} \hat{V}^{\pi}$
      Replace $\pi_{\theta^{S_2}} = \pi_{\theta^*}$ and update m' using PID controller
    end if
  end for
  if $\hat{V}^{\pi_{\theta^*}} > \hat{V}^{\pi_{\theta^T}}$ **then**
    Update $\pi_{\theta^T} = \pi_{\theta^*}$
  end if
end while

D. Entropy-Regulated Adversarial Training

Conventional adversarial training in RL often suffers from perturbation overfitting, where agents converge to defenses tailored to specific attack patterns, limiting adaptability to novel threats. This brittleness stems from static perturbation generation that fails to diversify adversarial exposures, ultimately compromising generalizable robustness. To enhance generalization against diverse attacks, HORP introduces a novel entropy regulation mechanism that dynamically injects uncertainty into adversarial perturbations, balancing exploitation of known attacks with exploration of new perturbation directions.

In standard adversarial training following established methods [4], [5], an adversary ν observes the true state s_t and generates a perturbed state $\hat{s}_t$ by first computing parameters $\boldsymbol{\mu}_t$ and $\boldsymbol{\Sigma}_t$ from s_t, then sampling from a Gaussian distribution

$$\hat{s}_t \sim \mathcal{N}\left(\boldsymbol{\mu}_t, \boldsymbol{\Sigma}_t\right). \tag{30}$$

However, as training progresses, perturbations tend to become increasingly deterministic, reducing entropy and limiting exploration. To counteract this entropy collapse, we introduce our novel entropy regulation mechanism through adaptive uncertainty injection.

Our method begins by quantifying perturbation magnitude using the squared Euclidean distance

$$d_t = \|s_t - \hat{s}_t\|_2^2. \tag{31}$$

We then compute a weighted moving average to capture recent perturbation trends

$$\bar{d}_t = \beta\bar{d}_{t-1} + (1-\beta)\, d_t \tag{32}$$

where $\beta \in [0, 1)$ is a momentum factor and $\bar{d}_0$ is initialized appropriately to zero.

Using $\bar{d}_t$, we define an adaptive scaling factor α_t to adjust uncertainty levels

$$\alpha_t = \sigma\left(\bar{d}_t\right) + b \tag{33}$$

where $\sigma(\cdot)$ is the sigmoid function and $b \geq 0$ is a bias factor. The bias b ensures baseline exploration, particularly important during early training when the adversary cannot yet generate effective perturbations. This design follows established RL methods [50] to maintain minimal entropy injection. The value of b is typically set to a small positive constant (e.g., $b = 0.1$) to mitigate exploration collapse while avoiding excessive noise that could impede learning.

To generate diverse perturbations, we combine two sampling strategies.

1) *Exploitation Samples*: Standard adversarial perturbations $\tilde{s}_t^{(i)} \sim \mathcal{N}(\boldsymbol{\mu}_t, \boldsymbol{\Sigma}_t)$ for $i = 1, 2, \ldots, m_1$.
2) *Exploration Samples*: Entropy-enhanced perturbations where $z^{(i)} \sim \mathcal{U}(-\alpha_t, \alpha_t)$ and $\tilde{s}_t^{(i)} \sim \mathcal{N}(\boldsymbol{\mu}_t + z^{(i)}, \boldsymbol{\Sigma}_t)$ for $i = m_1 + 1, \ldots, n$.

This yields a diverse candidate set $\{\tilde{s}_t^{(i)}\}_{i=1}^{n}$ with n total samples.

The core innovation is the optimization-based selection from this candidate set. We define a probability distribution P over $\{\tilde{s}_t^{(i)}\}$ that maximizes expected adversarial effectiveness while maintaining diversity. Formally, we solve

$$\begin{aligned} &\max_{P\in\Delta_{[n]}} \ \mathbb{E}_{\tilde{s}\sim P}\left[\nu\left(\tilde{s} \mid s_t\right)\right] \\ &\text{s.t. } D_{\text{TV}}\left(P \,\|\, U\right) \leq \eta \end{aligned} \tag{34}$$

where $\nu(\tilde{s} \mid s_t)$ is the adversary's likelihood of choosing perturbation $\tilde{s}$ given state s_t, $\Delta_{[n]}$ is the probability simplex over n elements, U is the uniform distribution over $\{\tilde{s}_t^{(i)}\}$, η controls the allowable deviation from uniformity, and $D_{\text{TV}}(P \,\|\, U) = \frac{1}{2}\sum_{i=1}^{n} |P_i - U_i|$ is the total variation distance.

This optimization problem is a linear program (LP). Solving it requires computing $\nu(\tilde{s}_t^{(i)} \mid s_t)$ for all n candidates using the Gaussian probability density function based on $(\boldsymbol{\mu}_t, \boldsymbol{\Sigma}_t)$, which requires $O(n \cdot d_s^2)$ operations, where d_s is the state dimension. The LP can be solved to optimality with polynomial complexity $O(n^{3.5})$ using interior point methods. For practical efficiency during training, we employ an approximate sorting-based method that first sorts candidates by descending likelihood values, then applies a water filling strategy to adjust probabilities until the TV-distance constraint is met, requiring only $O(n \log n)$ operations while preserving diversity benefits.

Proposition 1 (Closed-form Solution for Two Candidate Perturbations): For the entropy-constrained adversarial selection problem defined as

$$\max_{P\in\Delta_{[2]}} \; P_1\,\nu\left(\tilde{s}_t^{(1)} \mid s_t\right) + P_2\,\nu\left(\tilde{s}_t^{(2)} \mid s_t\right)$$
$$\text{s.t. } D_{\mathrm{TV}}(P \,\|\, U) \le \eta$$

the optimal distribution is given by

$$P^* = \left(\tfrac{1}{2} + \eta\,\mathrm{sgn}(\Delta\nu),\; \tfrac{1}{2} - \eta\,\mathrm{sgn}(\Delta\nu)\right) \tag{35}$$

where $\Delta\nu = \nu(\tilde{s}_t^{(1)} \mid s_t) - \nu(\tilde{s}_t^{(2)} \mid s_t)$, and $\mathrm{sgn}(\cdot)$ is the sign function. The proof is given in Section S2.10 of the supplementary material.

Algorithm 2 Entropy-Regulated Adversarial Training

Input: State s_t, adversarial policy ν, momentum coefficient β, bias term b, sample sizes m_1, n
Extract distribution parameters $(\boldsymbol{\mu}_t, \boldsymbol{\Sigma}_t) = \nu(\cdot|s_t)$
Sample $s_t^{\mathrm{adv}} \sim \mathcal{N}(\boldsymbol{\mu}_t, \boldsymbol{\Sigma}_t)$
Compute $d_t = \|s_t - s_t^{\mathrm{adv}}\|_2^2$
Update $\bar{d}_t = \beta\bar{d}_{t-1} + (1-\beta)d_t$
Calculate $\alpha_t = \sigma(\bar{d}_t) + b$
Generate exploitation samples: $\{\tilde{s}_t^{(i)}\}_{i=1}^{m_1} \sim \mathcal{N}(\boldsymbol{\mu}_t, \boldsymbol{\Sigma}_t)$
for $i = m_1 + 1$ **to** n **do**
  Sample $z^{(i)} \sim \mathcal{U}(-\alpha_t, \alpha_t)$
  Generate exploration sample $\tilde{s}_t^{(i)} \sim \mathcal{N}(\boldsymbol{\mu}_t + z^{(i)}, \boldsymbol{\Sigma}_t)$
end for
Determine optimal distribution P^* over $\{\tilde{s}_t^{(i)}\}_{i=1}^{n}$
Sample $\hat{s}_t \sim P^*$
Return: Perturbed state $\hat{s}_t$

The final perturbation $\hat{s}_t$ is sampled from the optimized distribution P^*. By maintaining diverse adversarial exposures throughout training, this entropy-regulated method enhances generalization capabilities against various attack types, contributing to improved robustness. The complete procedure is summarized in Algorithm 2.

E. Computational Complexity Analysis

This section provides a comprehensive computational complexity analysis of HORP, examining both time and space requirements to evaluate the method's scalability. We define key parameters for clarity: Φ denotes the number of parameters in a single policy or value network (all networks are of comparable size), with a single forward pass costing $O(\Phi)$; C_{env} represents the cost of one environment step; C_{bound} is the cost factor for convex relaxation-based bound computation per state; K_Q indicates the number of gradient ascent steps for the $\max Q$ operation; n is the number of candidate perturbations generated per step; and m is the fixed policy replacement interval. Additionally, T_{eval} denotes steps used for policy evaluation, while d_s and d_a are state and action space dimensions, respectively.

1) Time Complexity: We analyze the time complexity of each core HORP component.

1) *Policy-Guided Learning (PGL):* For each state, computing action bounds $[\underline{a}, \overline{a}]$ via convex relaxation costs $O(C_{\mathrm{bound}} \cdot \Phi)$, while solving $\max_{\hat{a}} Q^{\pi_{\theta^S}}(s, \hat{a})$ through K_Q gradient ascent steps costs $O(K_Q \cdot \Phi)$ per state. Policy distribution divergence computation requires $O(d_a^2)$ per state, and guide reward with value function updates cost $O(T)$ and $O(T \cdot \Phi)$, respectively, yielding overall PGL complexity of $T_{\mathrm{PGL}} = O(T \cdot (C_{\mathrm{bound}} + K_Q) \cdot \Phi + T \cdot d_a^2)$. Given that Φ (encompassing network depth and width) typically dominates d_a^2, complexity is effectively linear in T and Φ, with C_{bound} and K_Q determining overhead.
2) *Adaptive Performance-Aware Policy Optimization (APPO):* The PID controller incurs negligible $O(1)$ cost. At the same time, periodic policy evaluation involves rolling out 4 policies (best student and three students) for T_{eval} steps at cost $O(T_{\mathrm{eval}} \cdot (\Phi + C_{\mathrm{env}}))$, resulting in amortized complexity over m training steps of $T_{\mathrm{APPO}} = O((T_{\mathrm{eval}})/m \cdot (\Phi + C_{\mathrm{env}}))$.
3) *Entropy-Regulated Adversarial Training (ERAT):* For each of T environment interactions, ERAT generates n candidate perturbed states costing $O(\Phi + n \cdot d_s)$, evaluates adversary likelihood per candidate at $O(n \cdot d_s^2)$ cost, and performs optimization-based selection with $O(n \log n)$ complexity, leading to per-step complexity of $O(\Phi + n \cdot (d_s + d_s^2 + \log n))$ and total complexity of $T_{\mathrm{ERAT}} = O(T \cdot (\Phi + n \cdot (d_s + d_s^2 + \log n)))$. Thus, T_{ERAT} scales linearly with T, n, and Φ, and linearly to quadratically with d_s.

Incorporating base environment interaction costs $O(3T \cdot (\Phi + C_{\mathrm{env}}))$ for three student policies, the overall time complexity per iteration is

$$T_{\mathrm{total}} = O\left(3T\left[\left(C_{\mathrm{bound}} + K_Q + \frac{T_{\mathrm{eval}}}{mT} + 2\right)\Phi + \left(1 + \frac{T_{\mathrm{eval}}}{mT}\right)C_{\mathrm{env}} + d_a^2 + n\left(d_s + d_s^2 + \log n\right)\right]\right). \tag{36}$$

Complexity scales linearly with student policy count, trajectory length T, and network parameters Φ. For state–action dimensions, scaling is linear in d_a and linear to quadratic in d_s. In practice, environment interaction costs $3T \cdot (\Phi + C_{\mathrm{env}})$ dominate, as C_{env} typically exceeds neural network operations by orders of magnitude. The factor of 3 stems from concurrent environment interaction by three student policies, a key design choice enhancing exploration and robustness.

2) Space Complexity: Memory requirements are analyzed componentwise.

1) *PGL:* Persistent memory includes parameters for one student policy network, one teacher policy network, one adversary network, and student's associated value function networks (environment value, guide value, and Q-network), totaling $O(6\Phi)$, while the experience buffer storing T tuples $(s_t, \hat{s}_t, a_t, r_t, s_{t+1})$ requires $O(T \cdot (3d_s + d_a))$ space, resulting in $S_{\mathrm{PGL}} = O(T \cdot (3d_s + d_a) + 6\Phi)$.
2) *APPO:* Policy evaluation requires transient loading of policy parameters costing $O(\Phi)$, and the PID controller uses only $O(1)$ space for performance gap storage, with no significant persistent memory beyond S_{PGL} allocated.
3) *ERAT:* Per-step processing requires $O(n \cdot d_s)$ space for candidate perturbations and likelihoods, which is

released after perturbation selection and does not contribute to persistent footprint.

Total persistent space complexity is dominated by experience buffers and network parameters. HORP maintains parameters for three student policy networks (each with environment value, guide value, and Q-network), one teacher policy network, one best student policy network, and one adversary network, totaling $O(15\Phi)$. Combined with experience buffers

$$S_{\text{total}} = O\left(T \cdot (3d_s + d_a) + 15\Phi\right). \tag{37}$$

Memory footprint scales linearly with trajectory length T and state–action dimensions.

3) Comparison With Baselines: We compare HORP's computational profile with established robust RL methods. ATLA-PPO [4] alternates training between single policy and adversary, achieving $O(T \cdot (2\Phi + C_{\text{env}}))$ time and $O(T \cdot (3d_s + d_a) + 4\Phi)$ space complexity, offering computational efficiency but limited robustness. PROTECTED-PPO [10] maintains an ensemble of E policies, resulting in higher complexities of $O(T \cdot (E + 1) \cdot (2\Phi + C_{\text{env}}))$ time and $O((E + 3)\Phi + T \cdot (3d_s + d_a))$ space, due to extensive per-policy environment interactions. Although HORP introduces PGL overhead, efficient convex relaxation methods compute action bounds with constant-factor overhead relative to direct policy computations. The resulting tight bounds ensure max-Q operation remains computationally tractable. Consequently, HORP achieves a superior robustness–efficiency trade-off unlike PROTECTED-PPO, whose policy count grows linearly with E; it maintains a constant policy count while providing enhanced robustness over ATLA-PPO via multipolicy guidance rather than single-policy adversarial training.

IV. Experiments and Analysis

In this section, we conduct comprehensive experiments to evaluate the HORP method, addressing five questions: 1) Does HORP outperform existing methods in both natural performance and adversarial robustness? 2) Can HORP maintain robustness against dynamic attacks? 3) What are HORP's sample complexity and computational overhead? 4) How sensitive is HORP to the initial teacher policy? and 5) How do HORP's individual components contribute to overall performance? All questions are addressed through extensive experiments.

A. Experimental Setup

We evaluate HORP across four MuJoCo environments with continuous action spaces: Hopper, Walker2d, Halfcheetah, and Ant. We benchmark HORP against several state-of-the-art robust training methods, including ATLA-PPO [4], PA-ATLA-PPO [5], WocaR-PPO [3], PROTECTED-PPO [10], and ACoE-PPO [28], utilizing the pretrained models provided by the respective authors. To assess performance, we report natural rewards to represent scenarios without any attacks and evaluate the robustness of agents using five attack methods, varying in strength: 1) random (random bounded perturbation); 2) robust sarsa (RS) [37] (attacking with a robust action–value function); 3) MAD [37] (maximal action difference); 4) SA-RL [4] (identifying the optimal state adversary); and 5) PA-AD [5] (an enhanced version of SA-RL with improved attack efficiency).

For implementation and hyperparameter configurations, we use a single-layer LSTM with 64 hidden neurons for the Halfcheetah environment and retain the original fully connected MLP architecture for Hopper, Walker2d, and Ant. Training is conducted for 10 million steps (4882 iterations) in Ant and 5 million steps (2441 iterations) in the other three environments. Both agents and adversaries are optimized independently using the PPO method during the adversarial training process. To ensure fair comparison, we conduct extensive hyperparameter searches for adversary configurations. For RS attacks, we evaluate 30 configurations spanning the following parameter ranges.

1) *Perturbation Bound:* $\epsilon_a \in \{0.02, 0.05, 0.1, 0.15, 0.2, 0.3\}$
2) *Regularization Coefficient:* $\lambda_r \in \{0.1, 0.3, 1.0, 3.0, 10.0\}$

For the RL-based adversarial attacks (SA-RL and PA-AD), we conduct an extensive grid search over 108 configurations with the following parameter spaces.

1) *Adversary Policy Learning Rate:* $\alpha_p \in \{10^{-3}, 3 \times 10^{-3}, 10^{-2}\}$
2) *Value Network Learning Rate:* $\alpha_v \in \{10^{-3}, 3 \times 10^{-3}, 10^{-2}\}$
3) *PPO Clipping Parameter:* $\epsilon_c \in \{0.2, 0.4\}$
4) *Entropy Regularization coefficient:* $\beta_e \in \{0.0, 10^{-5}, 3 \times 10^{-5}, 10^{-4}, 3 \times 10^{-4}, 10^{-3}\}$

For all attack methods, the optimal adversary configuration yielding the lowest agent reward was selected for final evaluation. All experiments utilize NVIDIA GeForce RTX 3090 GPU with Python 3.8.13.

B. Evaluation of HORP-PPO

The performance of HORP under natural and adversarial conditions is summarized in Table I, reporting mean rewards and standard deviations over 50 evaluation episodes across 10 random seeds (0–9). To address statistical reliability concerns, we provide 95% confidence intervals and significance tests comparing HORP against each baseline in Section S3.2 of the supplementary material. As shown in Table I, HORP achieves superior natural rewards in Hopper, Walker2d, and Halfcheetah, while PROTECTED-PPO obtains higher natural performance in Ant (5884 ± 102 versus 5634 ± 28, $p < 0.05$), illustrating the trade-off between nominal performance and adversarial robustness in high-dimensional tasks. Under adversarial settings, HORP demonstrates strong worst-case performance: it excels across most attacks in Hopper and shows statistically significant improvements in Walker2d. In Halfcheetah, HORP performs best under Random, SA-RL, and PA-AD attacks but is outperformed by PA-ATLA-PPO and PROTECTED-PPO under RS and MAD attacks, highlighting the complementary nature of robustness strategies. For Ant, HORP maintains competitive adversarial performance despite lower natural rewards, with significant improvements under SA-RL and PA-AD attacks.

TABLE I

AVERAGE EPISODE REWARDS ± STANDARD DEVIATION OVER 50 EPISODES FOR FIVE BASELINES ACROSS FOUR ENVIRONMENTS, WITH EACH RESULT OBTAINED USING 10 RANDOM SEEDS (0–9). THE ATTACK BUDGET ϵ MATCHES THOSE USED IN RELATED STUDIES. THE BEST PERFORMANCES IN EACH EVALUATION METRIC ARE HIGHLIGHTED IN BOLD. IN EACH ROW, THE LOWEST ATTACK REWARD AMONG THE FIVE ATTACKS IS INDICATED AS THE BEST REWARD, AND GRAY ROWS HIGHLIGHT THE MOST ROBUST AGENTS

Environment	Model	Natural Reward	Attack Reward					Best Attack
			Random	RS	MAD	SA-RL	PA-AD	
	ATLA-PPO	3301 ± 540	3236 ± 549	2362 ± 817	3011 ± 706	1883 ± 807	2101 ± 799	1883 ± 807
Hopper	PA-ATLA-PPO	3608 ± 27	3461 ± 361	1488 ± 659	2984 ± 776	1674 ± 743	1804 ± 773	1488 ± 659
state-dim: 11	WocaR-PPO	3625 ± 98	3621 ± 73	1881 ± 601	3011 ± 701	1717 ± 613	2462 ± 859	1717 ± 613
ϵ=0.075	PROTECTED-PPO	3669 ± 36	3629 ± 31	2512 ± 786	2908 ± 694	2037 ± 517	2183 ± 761	2037 ± 517
	ACoE-PPO	3707 ± 32	3646 ± 38	2630 ± 897	3057 ± 822	1949 ± 608	2022 ± 811	1949 ± 608
	HORP-PPO (Ours)	**3756 ± 30**	**3728 ± 180**	**3481 ± 267**	**3206 ± 716**	**3191 ± 694**	**2698 ± 913**	**2698 ± 913**
	ATLA-PPO	3841 ± 474	3914 ± 357	3257 ± 950	3835 ± 491	3613 ± 702	3564 ± 759	3257 ± 950
Walker2d	PA-ATLA-PPO	5124 ± 1212	4975 ± 1242	3910 ± 1571	5155 ± 1151	3752 ± 2096	3985 ± 1606	3752 ± 2096
state-dim: 17	WocaR-PPO	4113 ± 1062	4471 ± 728	2820 ± 988	4293 ± 907	2957 ± 921	2690 ± 1544	2690 ± 1544
ϵ=0.05	PROTECTED-PPO	6310 ± 34	6306 ± 53	5702 ± 1365	6315 ± 46	5977 ± 1059	5919 ± 1069	5702 ± 1365
	ACoE-PPO	5345 ± 566	5231 ± 1256	5558 ± 854	5444 ± 335	5353 ± 1240	4886 ± 1512	4886 ± 1512
	HORP-PPO (Ours)	**6974 ± 254**	**7001 ± 121**	**6806 ± 114**	**6949 ± 333**	**6486 ± 703**	**6431 ± 959**	**6431 ± 959**
	ATLA-PPO	6335 ± 189	6220 ± 154	4839 ± 790	5729 ± 384	5050 ± 785	4900 ± 1148	4839 ± 790
Halfcheetah	PA-ATLA-PPO	6266 ± 163	6171 ± 138	5474 ± 911	5813 ± 131	4427 ± 709	4943 ± 310	4427 ± 709
state-dim: 17	WocaR-PPO	6210 ± 807	6132 ± 158	3120 ± 1778	**5924 ± 160**	3170 ± 1485	4033 ± 1499	3120 ± 1778
ϵ=0.15	PROTECTED-PPO	7032 ± 138	6429 ± 195	**5542 ± 252**	5809 ± 156	5130 ± 63	5466 ± 90	5130 ± 63
	ACoE-PPO	6367 ± 187	6255 ± 171	5022 ± 1506	5196 ± 70	4929 ± 88	4841 ± 162	4841 ± 162
	HORP-PPO (Ours)	**7616 ± 162**	**7038 ± 219**	5136 ± 164	5321 ± 203	**5144 ± 85**	**5586 ± 316**	**5136 ± 164**
	ATLA-PPO	5239 ± 408	5182 ± 179	3248 ± 147	5252 ± 153	3459 ± 141	2210 ± 74	2210 ± 74
Ant	PA-ATLA-PPO	5054 ± 641	5068 ± 146	3115 ± 1619	4815 ± 745	3561 ± 1318	2963 ± 1652	2963 ± 1652
state-dim: 111	WocaR-PPO	5567 ± 118	5518 ± 142	3200 ± 1667	5388 ± 123	2083 ± 50	2229 ± 80	2083 ± 50
ϵ=0.15	PROTECTED-PPO	**5884 ± 357**	5302 ± 830	4249 ± 408	4745 ± 120	4297 ± 66	4122 ± 79	4122 ± 79
	ACoE-PPO	5526 ± 128	5257 ± 95	3779 ± 1910	5171 ± 137	4017 ± 77	3328 ± 141	3328 ± 141
	HORP-PPO (Ours)	5634 ± 100	**5549 ± 124**	**5116 ± 868**	**5398 ± 146**	**4975 ± 127**	**4753 ± 1244**	**4753 ± 1244**

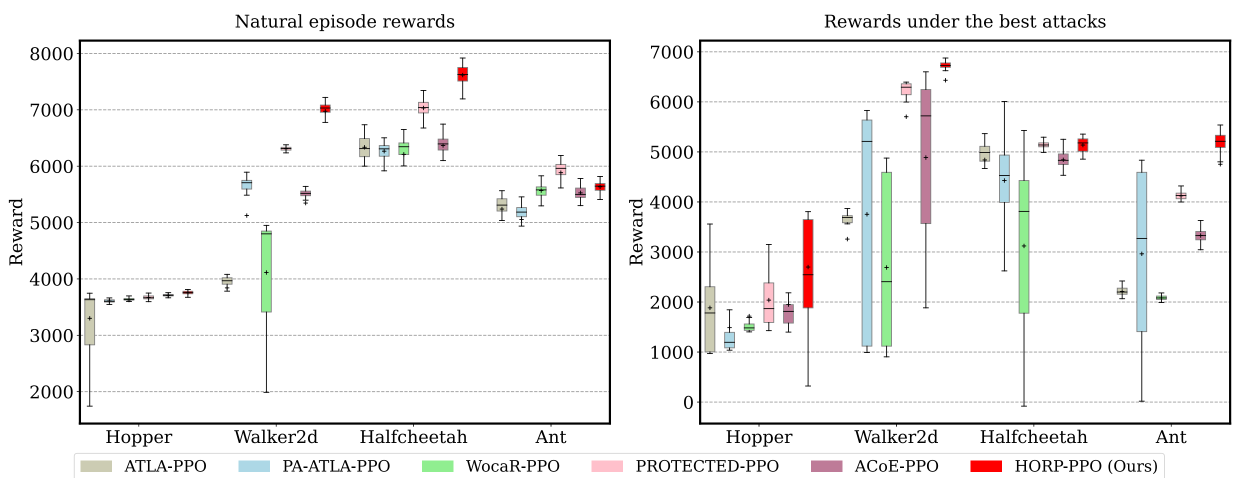

Fig. 2. Performance distribution analysis of PPO variants under natural and adversarial conditions across four experimental domains. Box plots summarize reward statistics over 50 episodic trials per run, with results aggregated across 10 random seeds (0–9). Box boundaries mark interquartile ranges (25th–75th percentiles), central crosses denote median values, and whiskers extend to extreme observations.

We observe that HORP exhibits higher variance in certain settings (e.g., PA-AD attack in Hopper: 2698 ± 913; Ant: 4753 ± 1244). This behavior aligns with our entropy-regulated training strategy, which deliberately injects controlled uncertainty to enhance generalization by exploring diverse perturbation directions. While this design choice increases performance

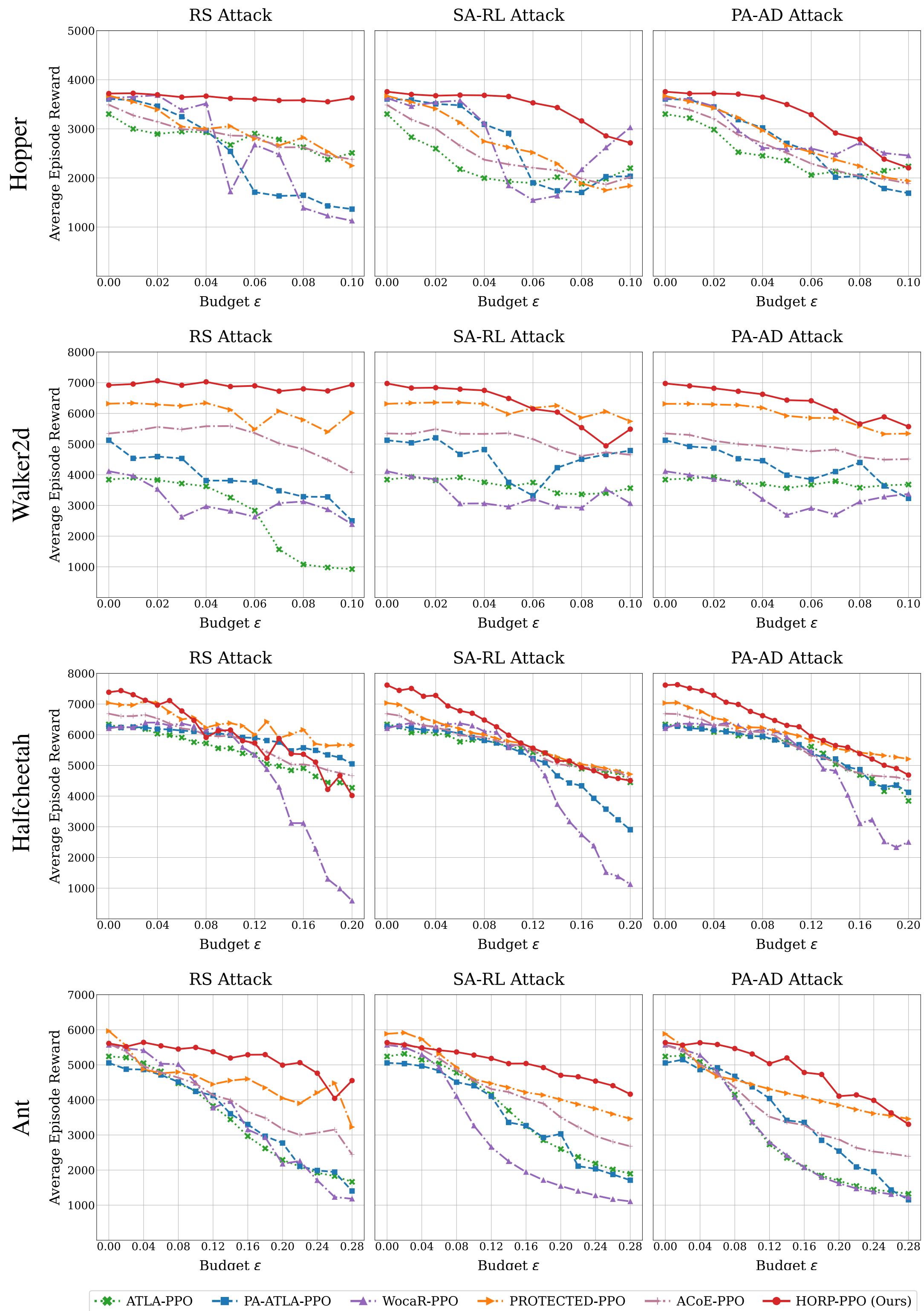

Fig. 3. Evaluating the robustness of PPO agents under varying budget constraints ε across three distinct attack methods in four different environments, where state perturbations adhere to $\|s - \hat{s}\|_\infty \leq \varepsilon$. Each point on this graph represents the mean outcome over 50 episodes, averaged across 10 random seeds (0–9).

variance, it fundamentally prevents overfitting to specific attack patterns and enhances worst-case robustness, as evidenced by HORP's competitive best attack metrics across all environments. Fig. 2 further illustrates these trends, showing reward distributions under the strongest attacks and natural conditions. Despite higher variance in some cases, HORP maintains superior central tendency under attack compared to the severe degradation observed in baselines, reinforcing

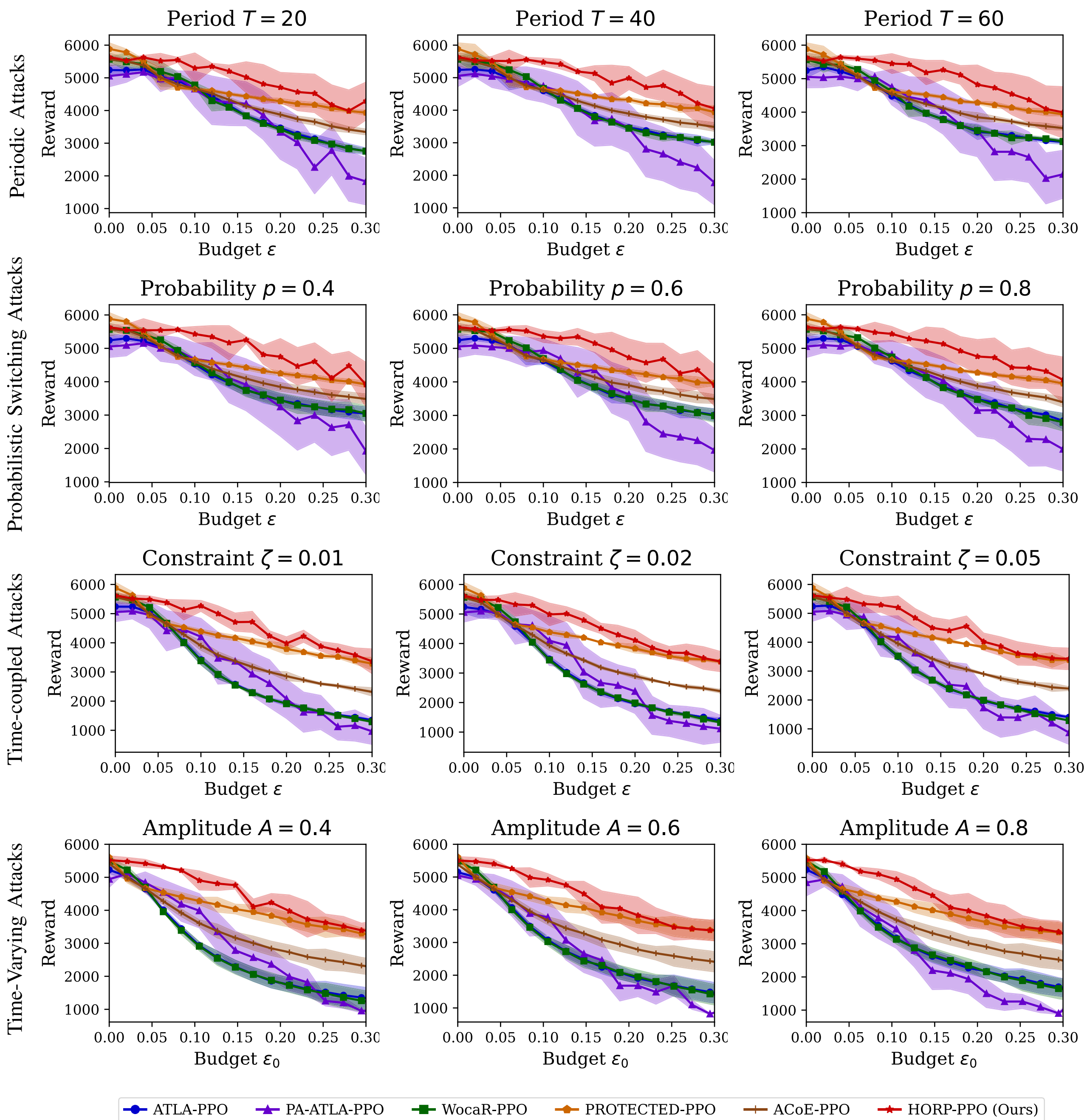

Fig. 4. Shaded regions represent half standard deviation around mean performance across 50 trials per configuration over 10 random seeds (0–9).

its ability to sustain policy effectiveness against strong adversaries. Additionally, Fig. 3 demonstrates HORP's superiority across most attack intensities in all environments. Notably, PROTECTED-PPO's requirement for maintaining multiple policy networks and performing extensive adjustments tailored to specific attackers may limit its practical applicability, particularly in safety-critical scenarios where computational efficiency and deployment simplicity are crucial.

C. Performance Against Dynamic Attacks

This section evaluates HORP against four types of nonstationary attacks in the Ant environment, all derived from the strong PA-AD adversary, to ensure challenging assessment scenarios.

Periodic attacks alternate between benign phases (no attack) and aggressive phases (PA-AD attack) with periods $T \in \{20, 40, 60\}$ time steps, testing resilience to intermittent threats.

Probabilistic switching attacks randomly transition between dormant (no attack) and active (PA-AD attack) states with probabilities $p \in \{0.4, 0.6, 0.8\}$ at 20-step intervals, evaluating robustness under stochastic attack patterns.

Time-coupled attacks impose temporal smoothness constraints $\|\hat{s}_t - \hat{s}_{t-1}\|_\infty \leq \zeta$ with $\zeta \in \{0.01, 0.02, 0.05\}$, ensuring

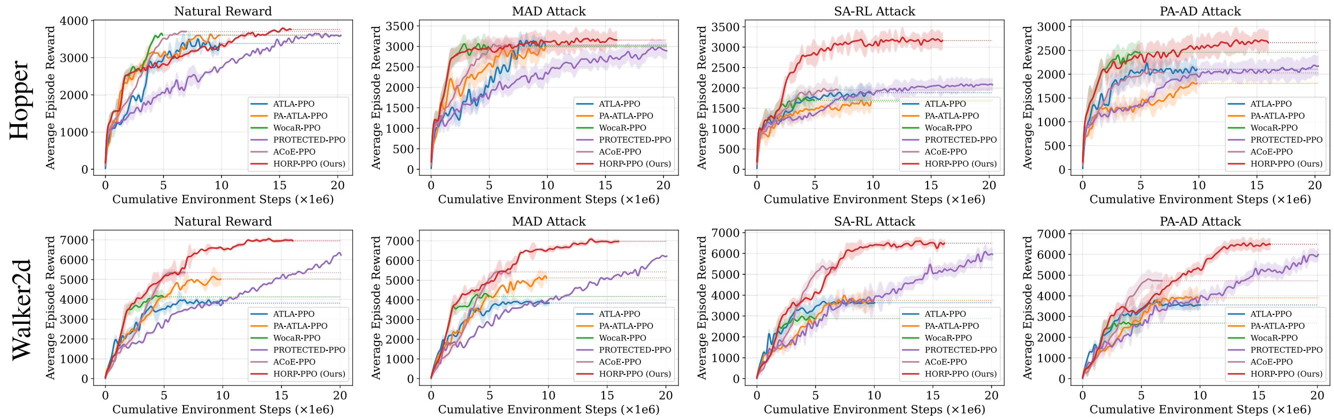

Fig. 5. Performance progression of PPO variants under natural and adversarial conditions across Hopper and Walker2d environments. Each subplot illustrates the relationship between environmental interaction steps and policy performance. Shaded regions denote 95% confidence intervals computed over 50 evaluation episodes across 10 random seeds (0–9).

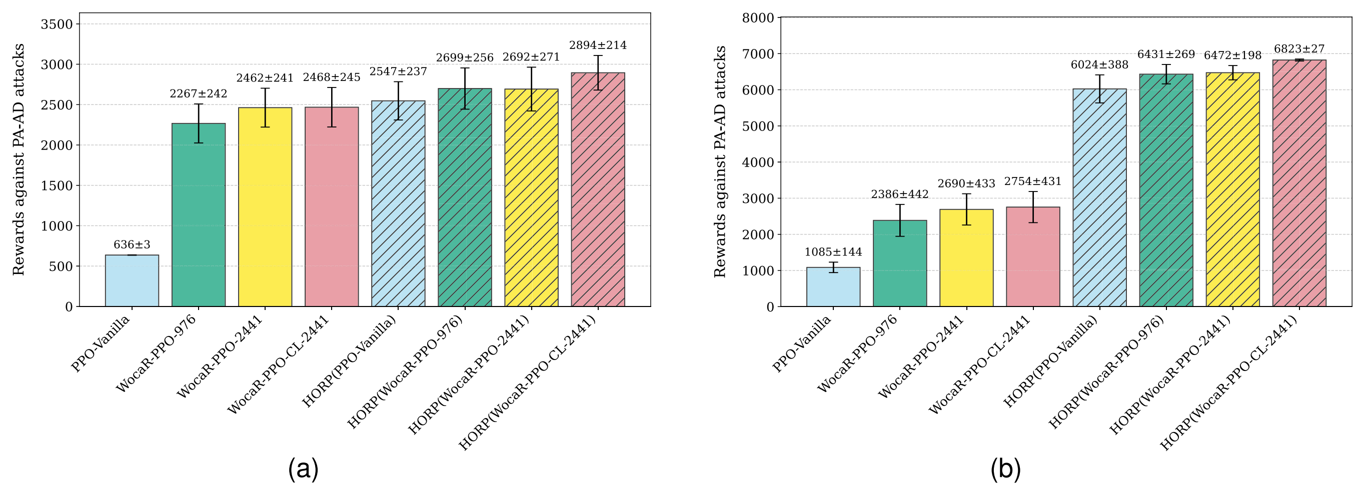

Fig. 6. Sensitivity analysis comparing different initial teacher policies and their HORP-enhanced counterparts under PA-AD attack in Hopper and Walker2d environments. Solid bars represent initial teacher policies; hatched bars show performance after HORP enhancement. Results display mean rewards with 95% confidence intervals over 50 evaluation episodes [10 random seeds (0–9)]. (a) Hopper. (b) Walker2d.

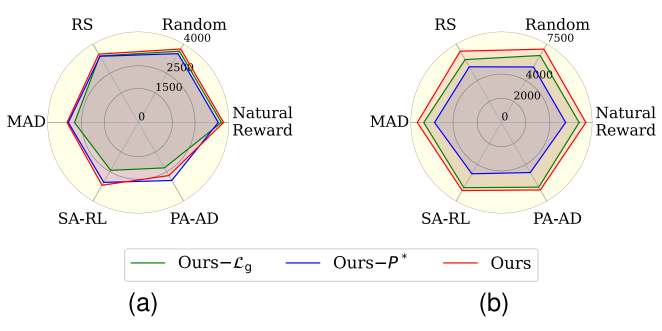

Fig. 7. Ablation performance of the guidance-enhanced policy loss $\mathcal{L}_g$ and the optimal perturbed states distribution P^* under natural conditions and various attacks in (a) Hopper and (b) Walker2d environments.

perturbations evolve gradually to simulate physical system constraints.

Time-varying attacks modulate perturbation intensity via $\varepsilon(t) = \varepsilon_0 + A \cdot \sin(2\pi t/T) + \eta(t)$ with amplitudes $A \in \{0.4, 0.6, 0.8\}$, period $T = 100$, base attack intensity ε_0, and Ornstein–Uhlenbeck noise $\eta(t)$, representing realistic nonstationary threat models with continuously varying intensity.

Fig. 4 shows that HORP maintains robust performance across most dynamic attack scenarios in the Ant environment, demonstrating effective adaptation to diverse temporal patterns and intensity modulations encountered in complex control environments.

D. *Efficiency Analysis*

Our efficiency analysis addresses computational requirements from both sample and time perspectives. Fig. 5 shows that HORP achieves competitive sample efficiency while attaining superior final performance across environments. PROTECTED-PPO exhibits the lowest sample efficiency due to maintaining multiple nondominated policy sets (with the ensemble size set to $E = 7$ following the original authors' recommendation for optimal performance), while WocaR-PPO and ACoE-PPO converge faster by avoiding RL-based adversary training, though at lower performance ceilings. HORP maintains comparable sample efficiency to

ATLA-PPO and PA-ATLA-PPO while delivering significantly better final performance, particularly against strong adversaries like SA-RL and PA-AD. In terms of computational overhead, HORP's training time is comparable to ATLA-PPO and PA-ATLA-PPO, notably faster than PROTECTED-PPO, though about twice as long as WocaR-PPO and ACoE-PPO. This additional computation is justified by substantial robustness gains. Complete training duration and GPU memory measurements are provided in Section S3.3 of the supplementary material.

HORP's favorable sample efficiency stems from its architectural design, where multiple student policies interact with the environment concurrently to enhance exploration diversity. The periodic reset mechanism further improves learning efficiency by enabling immediate knowledge transfer upon policy improvements. This coordinated method effectively reduces redundant exploration and accelerates collective policy optimization, justifying the computational investment through improved robustness and performance.

E. Sensitivity Analysis of Initial Teacher Policy

A potential consideration for our method is the sensitivity to the initial teacher policy, as the iterative policy optimization builds upon this foundation. To examine this aspect, we evaluate HORP under PA-AD attack across all four environments using different initial teacher policies. All previous experiments employed WocaR-PPO-976 (WocaR-PPO trained for 976 iterations with fixed attack budget ϵ_{target}) as the initial teacher. For this sensitivity analysis, we additionally consider three other initialization strategies: PPO-Vanilla (standard PPO trained for 2441 iterations), WocaR-PPO-2441 (WocaR-PPO trained for 2441 iterations with fixed attack budget ϵ_{target}), and WocaR-PPO-CL-2441 (curriculum learning with progressively increasing attack budget ϵ over 2441 iterations). The curriculum learning method mitigates the risks of catastrophic forgetting and performance collapse associated with premature exposure to strong state perturbations, thereby generating more reliable initial policies. All other experimental parameters remain the same.

Fig. 6 demonstrates that enhanced robustness and stability in the initial teacher policy consistently lead to higher rewards. This indicates that with more reliable initial guidance, HORP's policy guidance mechanism more effectively directs the optimization process toward near-optimal performance regions. Notably, even when initialized with the easily attainable standard PPO policy, our method surpasses all baseline methods in Table I, alleviating concerns about HORP's dependency on specialized initial teacher policies. Detailed sensitivity studies across all four MuJoCo environments are provided in Section S3.4 of the supplementary material.

F. Ablation Study

In this section, we present the ablation results for the guidance-enhanced policy loss $\mathcal{L}_{\text{g}}$ and the optimal perturbed states distribution P^{*} within the Hopper and Walker2d environments. Our experiments compare the performance of the original HORP-PPO method against variants excluding either $\mathcal{L}_{\text{g}}$ or P^{*}. Fig. 7 illustrates a notable decrease in performance when either component is omitted, confirming their vital roles in enhancing the robustness of HORP-PPO.

V. Conclusion

In this article, we present HORP, an innovative method designed to enhance the robustness of RL policies against adversarial perturbations. HORP integrates a policy guidance mechanism that leverages historically optimal policies to direct robust policy optimization, ensuring efficient adaptation to adversarial challenges. The adaptive performance-aware optimizer dynamically recalibrates policy updates through real-time performance diagnostics, maintaining robustness improvements throughout training iterations. Simultaneously, entropy-regulated adversarial training systematically diversifies perturbations via controlled stochastic injections, enhancing generalization across diverse threat patterns.

Notwithstanding these contributions, an important limitation of HORP is its current focus on ℓ_p-bounded state perturbations, which aligns with standard adversarial RL benchmarks but omits practical threat models like semantic distortions (e.g., illumination variations) or localized patch attacks prevalent in vision-based autonomous systems. Future work will extend HORP's defense paradigm to address photometric distortions and spatial attacks in real-world vision-RL applications. Integrating unsupervised learning techniques may further enhance policy generalization across unseen environments, advancing the transition from theoretical robustness to practical deployment.

Availability of Code

Code is available at https://github.com/Register-for-HORP/HORP-RL.

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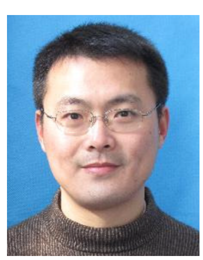

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